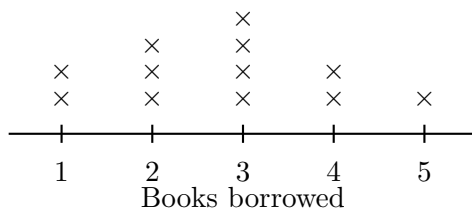


Find and Fix the Mistake

Diagnosing scale errors on line plots and attribute errors on shapes

Directions: In every problem another student has made a mistake. Part (a) asks you to fix the answer. Part (b) asks you to say what went wrong — write it in a sentence, in your own words. Use **Line Plot P** and **Shape Set Q** below.

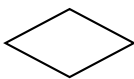
Line Plot P — Books borrowed by each student in one class



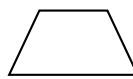
Shape Set Q



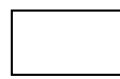
W



X



Y



Z

1. A student drew a scale under a line plot and labelled the five ticks 1, 2, 3, 5, 6. The steps are supposed to be the same size all the way along.

- (a) Which number belongs where the student wrote 5? _____
- (b) Explain why an uneven scale makes a line plot misleading.

2. Looking at Line Plot P, a student says “5 students borrowed 3 books,” because five numbers are printed along the scale.

- (a) How many students really borrowed 3 books? _____
- (b) Name the mistake the student made.

3. A student says shape X in Shape Set Q is a square, because it has four sides that are all the same length.

- (a) How many square corners does shape X really have? _____
- (b) Name the mistake the student made.

4. To find how many students are shown on Line Plot P, a student added the numbers along the scale and got 15.

- (a) How many students are really shown on the plot? _____
- (b) Name the mistake the student made.

5. A student found the difference between the greatest and the least number of books on Line Plot P by adding them, and got 6.

- (a) What is the correct difference? _____
- (b) Name the mistake the student made.

6. A student put shape Y from Shape Set Q into the group “has two pairs of parallel sides.”

- (a) How many pairs of parallel sides does shape Y really have? _____
- (b) Name the mistake the student made.

7. A student drew a scale from 1 to 5, counting by 1, but made the gap between two of the ticks twice as wide as the others.

- (a) On a correct scale from 1 to 5 counting by 1, how many equal gaps are there between the first tick and the last? _____
- (b) Explain why gaps of different widths make the finished plot misleading.

8. A student copied Line Plot P, which shows 12 students, but the copy has only 11 marks on it.

- (a) How many marks are missing from the copy? _____
- (b) Explain how to find which value was left out, without plotting the whole thing again.